

Medical Thermal Imaging Technologies for Breast Cancer and Tumor Detection

A Comprehensive Literature Review

Objective: To review and understand the available medical thermal imaging technologies and identify possible future research directions.

1. Introduction

Cancer remains one of the most significant global health burdens of the twenty-first century. According to global cancer statistics, breast cancer alone accounts for over 2.1 million new cases annually and is the leading cause of cancer-related mortality among women worldwide (Bray et al., 2024). The prognosis of cancer is profoundly influenced by the stage at which it is detected: research indicates that approximately 96% of cancers are treatable if diagnosed before metastasis occurs (Al Husaini et al., 2024). This underscores the paramount importance of early, accurate, and accessible detection technologies.

Conventional screening modalities such as mammography, magnetic resonance imaging (MRI), computed tomography (CT), and biopsy have long served as the gold standard in cancer diagnosis. However, these methods carry significant limitations including ionizing radiation exposure, high cost, patient discomfort, invasiveness, reduced sensitivity in individuals with dense breast tissue, and limited accessibility in low- and middle-income countries (LMIC). These drawbacks have catalyzed growing research interest in non-invasive, radiation-free, and cost-effective alternatives.

Medical thermal imaging, also known as infrared thermography or digital infrared thermal imaging (DITI), has emerged as a compelling complementary diagnostic modality. Rooted in the biophysical principle that malignant tumors exhibit heightened

metabolic activity and increased vascularization — leading to measurable surface temperature anomalies — thermography captures infrared radiation emitted by the body to generate thermal maps of tissues. First introduced as a breast cancer screening tool in 1956, thermography experienced a period of reduced clinical interest following a 1977 study questioning its reliability compared to other modalities. However, advances in infrared sensor technology, computational modeling, artificial intelligence (AI), and machine learning (ML) have dramatically revitalized this field over the past two decades (Arora et al., 2020).

This literature review surveys the current landscape of medical thermal imaging technologies applied to cancer and tumor detection. It critically examines the underlying biological and physical principles, the evolution and types of thermographic techniques, the integration of AI and deep learning in thermal image analysis, applications across various cancer types, current limitations, and possible future research directions. The review is structured to provide an undergraduate-level yet thorough understanding of both the technical and clinical dimensions of this rapidly evolving field.

2. Biological Basis of Thermal Imaging in Cancer Detection

2.1 Tumor Thermobiology

The foundation of thermal imaging as a diagnostic tool lies in the distinct thermobiological behavior of malignant tissues. Cancer cells are characterized by uncontrolled proliferation, elevated metabolic rates, and rapid cell division — all of which generate excess heat compared to surrounding normal tissues. The Pennes bioheat transfer equation, widely used in thermographic modeling, mathematically relates tissue temperature distribution to metabolic heat generation, blood perfusion, and thermal conductivity. In malignant tumors, both the metabolic heat generation rate and blood perfusion rate are substantially elevated relative to healthy tissue (Kandlikar et al., 2017).

A critical biological mechanism underlying thermographic cancer detection is angiogenesis — the formation of new blood vessels to supply the rapidly growing tumor with oxygen and nutrients. Tumors actively stimulate angiogenesis through the release

of vascular endothelial growth factors (VEGF), leading to a denser, more chaotic network of blood vessels in and around the lesion. This increased vascularity results in greater blood flow and higher local temperatures, which manifest as thermal asymmetries or hot spots on the surface of overlying skin. These temperature anomalies can be detectable by sensitive infrared cameras even before the tumor grows large enough to be palpable or visible on conventional imaging (Gonzalez-Hernandez et al., 2024).

Studies have confirmed that cancerous tissues consistently present with higher surface temperatures compared to contralateral healthy tissue, with thermal asymmetries often ranging from 1 to 3 degrees Celsius. This temperature differential forms the basis for identifying tumors non-invasively through surface temperature mapping. Importantly, since these vascular changes precede structural tumor development, thermography theoretically offers the capability for very early-stage detection — a key advantage over structural imaging modalities.

2.2 Temperature Signatures Across Cancer Types

Different cancers present distinct thermal signatures depending on their metabolic activity, vascularity, and anatomical location. Breast cancer typically produces well-defined thermal asymmetries visible on the anterior chest surface, with malignant areas appearing as regions of elevated temperature. Thyroid cancer causes increased intranodular vascularity that elevates skin surface temperatures over the anterior neck (Chantasartassamee et al., 2024). Skin cancers, particularly melanoma and basal cell carcinoma, produce localized thermal elevations directly detectable on the skin surface. Brain tumors and colorectal cancers have also been studied using thermographic approaches, though with greater technical complexity due to depth and anatomical barriers.

The thermal contrast between malignant and benign tissues is further influenced by several physiological factors including ambient temperature, patient positioning, hormonal fluctuations, and tissue density. Standardization of imaging protocols is therefore critical to ensure reproducible and clinically meaningful thermal measurements.

3. Medical Thermal Imaging Technologies

3.1 Passive (Static) Infrared Thermography

Passive or static thermography involves measuring the natural heat emission from the body surface without any external stimulation. In this approach, a patient is brought to thermal equilibrium in a controlled environment (typically maintained at 18–24 degrees Celsius), and thermal images are acquired using a calibrated infrared camera. The thermal camera detects infrared radiation in the 8–14 micrometer wavelength range and converts it into temperature maps, typically displayed as false-color images where warmer areas appear as brighter or redder tones.

Early-generation thermography relied on liquid crystal thermography (LCT), which required direct skin contact and offered limited sensitivity. Modern passive thermography employs high-resolution infrared cameras such as the FLIR SC-620, which captures images at 640 x 480 pixel resolution with thermal sensitivity below 0.05 degrees Celsius. These cameras are non-contact, fast, and capable of capturing whole-body thermal maps in seconds. Passive thermography forms the basis for most current clinical and research studies in breast cancer detection (Kandlikar et al., 2017).

A significant protocol advancement has been the prone patient positioning technique, in which the patient lies face-down through an opening in an imaging table, allowing the breast to hang freely. This positioning, combined with multi-view imaging from eight different angles, eliminates compression artifacts and allows more accurate thermal mapping of breast tissue from multiple perspectives (Gonzalez-Hernandez et al., 2020).

3.2 Dynamic (Active) Infrared Thermography

Dynamic thermography goes beyond static temperature mapping by introducing a controlled thermal stimulus — either heating or cooling — and then observing the thermal response of tissues over time. The rationale is that malignant tissues respond differently to thermal perturbations compared to healthy tissues due to differences in blood perfusion, thermal conductivity, and metabolic activity. Key parameters extracted

from dynamic thermographic sequences include the rate of temperature change, thermal recovery time constants, and thermal diffusivity.

Dynamic techniques have demonstrated improved contrast and sensitivity compared to static methods. In cooling-based protocols, the skin surface is briefly cooled using a fan or cold stimulus, and the rate at which temperature recovers to baseline is monitored. Cancerous regions, with their higher vascularization, typically exhibit faster thermal recovery. This dynamic information provides additional discriminatory power that static images alone cannot capture.

Several specialized dynamic techniques have been developed for clinical research. Lock-in thermography (LIT) uses a periodically modulated heat source and extracts amplitude and phase images using lock-in detection algorithms, significantly enhancing signal-to-noise ratio. Thermal wave imaging (TWI) applies pulsed heat stimulation and analyzes the propagation of thermal waves through tissue. Rotational breast thermography (RBT) acquires thermal images as the camera rotates around the patient, capturing three-dimensional thermal information. These advanced dynamic methods are reviewed in detail by recent literature focusing on their application to skin and breast cancer detection (Mangiacotti et al., 2024).

3.3 Infrared Thermal Modulation Endoscopy (ITME)

A novel and clinically significant development is Infrared Thermal Modulation Endoscopy (ITME), a label-free technology that adapts dynamic thermal modulation imaging for endoscopic use. Developed for internal tumor detection, ITME incorporates an endoscopic infrared imaging system with anti-reflection germanium lenses. By applying controlled tissue cooling and measuring the differential thermal response, ITME successfully localizes tumors based on their distinct heat signatures. Preclinical studies using a syngeneic rectal tumor mouse model demonstrated that early-stage tumors were successfully identified, with malignant tissues showing a higher rate of temperature change compared to surrounding normal tissues. ITME represents a significant step toward non-invasive internal cancer surveillance (Kim et al., 2024).

3.4 Induced Current Electro-Thermal Imaging (ICET)

Induced Current Electro-Thermal (ICET) imaging is an emerging active technique that applies magnetically induced electrical currents to breast tissue, exploiting the known difference in electrical conductivity between tumor tissue and healthy tissue — tumors have approximately 2.5 times higher electrical conductivity than normal breast tissue. The induced currents generate differential heat patterns that increase the temperature contrast on the imaging surface, improving the detectability of both surface and deep-seated tumors. Tanriverdi and Gencer (2024) demonstrated through numerical simulation and phantom experiments that ICET imaging improved tumor detection sensitivity by 4 millimeters, particularly for deep lesions that passive thermography may miss.

3.5 Smartphone-Based Thermal Imaging

An important recent development is the miniaturization of thermal imaging technology into portable, affordable devices. Smartphone-attachable thermal sensors such as the FLIR One Pro have enabled researchers and clinicians to conduct thermal imaging in resource-limited settings. A clinical validation study evaluated an AI-powered breast cancer pre-screening solution (BCPS) based on a smartphone with a thermal imaging sensor and found comparable performance to mammography in first-pass breast cancer screening (Enpress Publisher, 2024). This development is particularly relevant for low- and middle-income countries where access to traditional mammography infrastructure is limited, opening new possibilities for community-level cancer screening programs.

4. Image Processing and Artificial Intelligence in Thermal Image Analysis

4.1 The Computer-Aided Diagnosis (CAD) Pipeline

The clinical utility of thermography depends heavily on the accuracy and reproducibility of image interpretation. Human interpretation of thermographic images is inherently subjective and prone to inter-observer variability. Computer-aided diagnosis (CAD) systems have therefore become central to thermographic cancer detection research. A standard CAD pipeline for thermographic analysis comprises several

sequential stages: image acquisition, preprocessing, segmentation, feature extraction, feature selection, and classification.

Preprocessing involves noise reduction, contrast enhancement, and normalization of thermal images to account for environmental variability. Segmentation isolates the region of interest (e.g., breast tissue) from surrounding background. Feature extraction identifies discriminative characteristics of thermal patterns — these may be handcrafted features (texture descriptors, statistical moments, Gabor filters) or automatically learned by deep neural networks. Classification assigns the extracted features to malignant or benign categories using machine learning or deep learning classifiers (Mamani et al., 2025).

4.2 Traditional Machine Learning Approaches

Earlier CAD systems for thermographic cancer detection relied on traditional machine learning algorithms applied to handcrafted features. Common classifiers include Support Vector Machines (SVM), Random Forests, k-Nearest Neighbours (kNN), Naive Bayes, and Artificial Neural Networks (ANN). Feature extraction methods employed include Gray-Level Co-occurrence Matrix (GLCM) for texture analysis, Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and statistical features such as mean temperature, standard deviation, entropy, and thermal asymmetry indices.

These approaches demonstrated promising results in controlled settings. For example, ensemble methods combining Random Forest, Extra Trees, AdaBoost, and Gradient Boosting classifiers have been applied to breast thermogram classification with good accuracy. However, traditional ML approaches are constrained by their dependence on expert-designed features, which may fail to capture complex, non-linear thermal patterns characteristic of early-stage malignancies.

4.3 Deep Learning and Convolutional Neural Networks

The advent of deep learning has transformed the field of medical image analysis, and its application to thermographic cancer detection has accelerated significantly since 2020. Convolutional Neural Networks (CNNs) are the dominant deep learning

architecture for thermogram analysis. CNNs automatically learn hierarchical feature representations directly from raw image data, eliminating the need for manual feature engineering and enabling the capture of subtle and complex thermal patterns.

Pre-trained deep CNN architectures including VGG16, ResNet, DenseNet, InceptionV3, and EfficientNet have been applied to breast thermogram classification through transfer learning — a technique that leverages knowledge gained from large natural image datasets (such as ImageNet) and fine-tunes the network for the specific thermographic domain. This approach is particularly valuable given the relatively small size of available thermographic datasets. Al Husaini et al. (2024) applied InceptionV3, InceptionV4, and a modified InceptionMv4 architecture to real-time thermography video streaming for breast cancer classification, achieving competitive accuracy. A study using VGG16 with transfer learning on the DMR-IR database demonstrated strong classification performance between malignant and benign breast thermograms (Ahmed, 2024).

A recent study by Tewolde et al. (2025) proposed an advanced multi-CNN CAD system called Thermo-CAD specifically for early breast cancer detection using thermography, achieving high accuracy while reducing computational complexity compared to single-model approaches. Another significant work by Al Husaini et al. (2025) proposed a CNN framework enhanced with Particle Swarm Optimization (PSO) for hyperparameter tuning, demonstrating that optimizing learning rates, batch sizes, and network architecture significantly reduces false positive rates in thermographic breast cancer classification — a critical requirement for clinical acceptance.

4.4 AI in Thyroid Cancer Detection via Thermography

Beyond breast cancer, AI-enhanced thermography has been explored for thyroid malignancy detection. A prospective clinical study at Ramathibodi Hospital (Chantasartassamee et al., 2024) developed an AI diagnostic tool combining infrared thermal images of the anterior neck with convolutional neural networks to distinguish malignant from benign thyroid nodules. The study employed a compact tCam-mini thermal camera connected to an iPad and used a standardized cooling protocol — applying gel to the anterior neck before imaging — to enhance thermal contrast. The AI

model achieved clinically meaningful diagnostic performance, demonstrating the feasibility of thermography as a complementary tool to ultrasound-guided fine-needle aspiration biopsy (FNAB) in thyroid cancer workup. A CNN-based approach for classifying thyroid nodules from infrared thermographic images similarly achieved 91.11% accuracy (Vijay et al., 2022).

4.5 Physics-Based and Hybrid AI Models

A particularly innovative direction in thermal imaging research integrates physics-based bio-heat transfer modeling with AI-driven image analysis. Forward modeling uses the Pennes bioheat transfer equation to simulate the expected surface temperature distribution for a given tumor size, location, and metabolic activity. Inverse modeling works in reverse — it starts from measured surface temperatures and reconstructs the underlying tumor parameters (size, depth, metabolic heat generation) through optimization algorithms. Gutierrez et al. (2024) validated an inverse heat transfer modeling approach combining patient-specific 3D breast geometry with surface thermal measurements, accurately predicting all 25 tumor cases in their clinical study with maximum size errors below 5 millimeters. These physics-informed models generate synthetic training data that can augment limited clinical datasets, thereby improving AI model generalization.

5. Applications of Thermal Imaging Across Cancer Types

5.1 Breast Cancer

Breast cancer has received the most extensive investigation in thermographic research, driven by the high global incidence of the disease and the limitations of mammography in dense breast tissue. Thermography is non-invasive, painless, involves no radiation, and does not require breast compression — features that make it particularly attractive as a screening supplement or alternative in specific populations. Clinical studies have consistently shown that thermography detects physiological changes associated with developing tumors earlier than structural imaging can detect anatomical changes (Kandlikar et al., 2017).

A recent comprehensive review by Mamani et al. (2025) reported that thermal imaging achieves diagnostic accuracy between 97% and 100% in optimized models for breast cancer detection. Deep learning systems integrating multi-view thermograms with clinical data have further improved classification performance over single-modality models. However, the U.S. Food and Drug Administration (FDA) has maintained that thermography should not replace mammography as a primary breast cancer screening tool, citing the lack of large-scale prospective clinical trials (FDA, 2023). The current consensus positions thermography as a valuable adjunct to mammography rather than a standalone replacement.

The primary thermographic database used in breast cancer research is the Database for Research in Mastology with Infrared Images (DMR-IR), maintained by Antonio Pedro University Hospital in Brazil. While this database has enabled extensive algorithm development, over-reliance on a single dataset raises concerns about the generalizability of trained models to other populations and imaging devices (Tewolde et al., 2025).

5.2 Skin Cancer

Skin cancers — including melanoma, basal cell carcinoma, and squamous cell carcinoma — are amenable to thermographic detection due to their superficial location. Malignant skin lesions exhibit elevated metabolic activity and increased vascularity, producing localized thermal elevations directly measurable on the skin surface. Infrared thermography combined with AI algorithms has been explored as a contact-free, rapid assessment tool for skin cancer screening. The integration of AI in dermatology has accelerated broadly, with CNNs and Vision Transformers achieving dermatologist-level performance in skin cancer classification from both visible-light dermoscopy and thermal images (Wei et al., 2024).

Thermographic techniques such as thermal wave imaging have been applied to distinguish malignant melanoma from benign nevi based on differential thermal response to pulsed heating stimulation. These studies highlight the potential of active thermography to characterize lesion depth and vascular architecture — information critical for surgical planning.

5.3 Thyroid Cancer

Thyroid nodules are extremely common in the adult population, with prevalence rates of up to 65% on autopsy studies. While most are benign, distinguishing malignant from benign nodules currently requires invasive fine-needle aspiration biopsy with cytological analysis. Infrared thermography offers a non-invasive alternative for pre-operative risk stratification. The biological basis is that malignant thyroid nodules have increased intranodular vascularity, resulting in elevated surface temperatures over the anterior neck region. Several studies have validated this principle experimentally and clinically, confirming that thermographic images of the neck contain discriminative information about the malignant potential of thyroid nodules (Chantasartassamee et al., 2024). Hybrid multimodal models combining ultrasound and infrared thermal images have demonstrated improved classification accuracy over either modality alone (Zhang et al., 2023).

5.4 Brain and Rectal Tumors

The application of thermography to internal tumors represents a frontier in the field. Thermographic approaches to brain tumor detection have been explored intraoperatively, with thermal cameras mounted on haptic robots to map temperature distributions on the exposed brain surface during neurosurgery, aiding real-time delineation of tumor margins. The development of ITME (described in Section 3.3) has extended thermographic principles to the gastrointestinal tract, successfully detecting early-stage rectal tumors in animal models through endoscopic thermal modulation imaging. While these applications remain at the preclinical or early translational stage, they suggest that thermographic principles are not limited to superficial tissues.

6. Current Challenges and Limitations

6.1 Clinical Validation and Regulatory Acceptance

One of the most significant barriers to the widespread clinical adoption of thermographic cancer detection is the limited scale and quality of prospective clinical validation studies. Most published research relies on small, single-center datasets with limited demographic diversity. The FDA has specifically declined to approve

thermography as a primary breast cancer screening tool, citing insufficient evidence from large randomized controlled trials. The lack of standardized imaging protocols — including room temperature, camera specifications, patient preparation requirements, and image acquisition timing — further complicates cross-study comparisons and meta-analyses (Mamani et al., 2025).

6.2 Dataset Limitations and Generalizability

The field of thermographic AI suffers from a profound scarcity of large, publicly available, clinically validated thermal image datasets. Most deep learning models are trained and evaluated on the DMR-IR database, which contains images from a relatively homogeneous patient population. Models optimized for one dataset frequently fail to generalize to images obtained from different cameras, protocols, or patient populations. This over-reliance on a single data source severely limits the external validity and clinical applicability of published models (Tewolde et al., 2025). Additionally, class imbalance — where benign cases vastly outnumber malignant cases — can bias classifier performance toward the majority class if not properly addressed through data augmentation or balanced sampling strategies.

6.3 Physiological Confounders

Thermal imaging is sensitive to numerous physiological and environmental confounders that can introduce artifacts and reduce diagnostic accuracy. Factors such as ambient room temperature, air conditioning currents, patient skin temperature at the time of imaging, menstrual cycle phase, hormonal contraceptive use, recent exercise, caffeine consumption, and prior skin treatments can all significantly alter the thermal signature of breast tissue. Inter-individual variability in baseline body temperature further complicates the interpretation of thermal asymmetries. Rigorous patient preparation protocols and strictly controlled imaging environments are necessary to minimize these confounders, but adherence to such protocols is challenging in routine clinical practice (Ebsco Research, 2024).

6.4 Depth Limitation

A fundamental physical constraint of passive thermography is its sensitivity limitation for deep-seated tumors. Infrared cameras detect radiation from the skin

surface, and the thermal signal from a tumor attenuates with increasing depth due to the insulating properties of overlying tissue layers. Tumors located deeper than 2–3 centimeters below the skin surface may not generate sufficient surface temperature contrast for reliable detection by passive thermography alone. This depth limitation is particularly relevant for breast cancer in women with larger breast volumes. Active dynamic techniques and ICET imaging partially address this limitation by enhancing thermal contrast, but deep tumor detection remains an active area of research.

6.5 Model Interpretability

While deep learning models have achieved impressive classification accuracy in controlled research settings, their 'black-box' nature poses a significant challenge for clinical adoption. Clinicians require interpretable, explainable diagnostic outputs that provide not only a classification decision but also a clear visual indication of the thermal regions driving that decision. Explainable AI (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM), SHAP (SHapley Additive exPlanations), and attention maps are beginning to be applied to thermographic classification models to improve transparency, but robust clinical validation of these explainability methods is still lacking (Mamani et al., 2025).

7. Future Research Directions

7.1 Multimodal Fusion and Complementary Imaging

One of the most promising future directions is the development of multimodal diagnostic systems that fuse thermographic data with other imaging modalities and clinical information. Combining thermal images with mammographic, ultrasound, MRI, or clinical biomarker data has the potential to significantly improve diagnostic accuracy beyond what any single modality can achieve. Recent studies integrating thermal images with clinical data using multi-input CNNs have demonstrated accuracy improvements of 3–7% over single-modality models. Future research should focus on developing standardized fusion frameworks and evaluating multimodal systems in large prospective clinical trials across diverse patient populations.

7.2 Federated Learning for Privacy-Preserving Data Sharing

The data scarcity problem in thermographic AI could be partially addressed through federated learning — a distributed machine learning paradigm in which multiple institutions collaboratively train a shared AI model without sharing raw patient data. Each institution trains a local model on its own dataset, and only model parameters (not patient images) are shared and aggregated centrally. This approach enables the development of more generalizable AI models while respecting patient privacy and institutional data governance requirements. While federated learning applications in thermographic diagnostics remain limited — with only 1–3 published articles in the 2022–2023 period — this area is expected to grow significantly as the field matures (Piotrowski et al., 2025).

7.3 Standardization of Protocols and Public Datasets

Future progress in thermographic cancer detection requires the establishment of internationally recognized imaging protocols covering room conditions, patient preparation, camera calibration, image acquisition positioning, and data annotation standards. Regulatory bodies, professional medical societies, and research consortia should collaborate to develop and disseminate these standards. Simultaneously, large-scale multicenter data collection efforts are needed to build diverse, demographically representative public thermographic databases that can support the training and external validation of AI models across different populations. This is of particular importance for developing-world populations, including African populations, where thermal imaging could offer the most transformative impact due to limited mammography infrastructure.

7.4 Wearable and Continuous Thermal Monitoring

Emerging flexible infrared sensor technologies and miniaturized thermal imaging systems present the possibility of wearable thermal monitoring devices capable of continuously tracking skin surface temperature distributions over extended periods. Such devices could enable dynamic longitudinal tracking of thermal signatures, capturing not only static temperature maps but also temporal thermal patterns that may provide earlier cancer warning signals. Integration with smartphone applications and

cloud-based AI analysis platforms could make such monitoring accessible to patients outside clinical settings, supporting decentralized early cancer detection programs particularly relevant in rural and resource-limited environments.

7.5 Integration of Nanoparticle-Enhanced Thermography

An innovative research direction involves the use of magnetic nanoparticles as contrast agents to enhance thermal imaging sensitivity for deep tumor detection. Nanoparticles selectively accumulate in tumor tissue due to the enhanced permeability and retention (EPR) effect, and can be externally heated using alternating magnetic fields, producing a localized temperature increase detectable by infrared cameras. Levy et al. (2010) demonstrated this concept through theoretical simulation and in vitro validation, showing that nanoparticle-enhanced thermography could significantly improve the detectability of sub-centimeter deep tumors. Future clinical translation of this approach, combined with precision nanoparticle targeting strategies, represents a high-potential frontier for the field.

7.6 Explainable AI and Clinical Decision Support

For thermographic AI systems to achieve regulatory approval and clinical adoption, future research must prioritize the development of clinically validated explainability frameworks. XAI methods that produce heat maps or attention overlays on thermal images — highlighting the specific regions and features driving the classification decision — will be essential for building clinician trust and enabling meaningful human-AI collaboration in cancer screening. Future work should evaluate whether XAI-enhanced thermographic systems improve diagnostic confidence and accuracy among clinicians compared to AI-only or human-only interpretation.

7.7 Real-Time Video Thermography

Static single-frame thermal imaging captures a snapshot of the thermal state of tissue at one moment in time. Real-time thermographic video streaming captures dynamic thermal changes over time, providing richer temporal information about tumor vascularity and metabolic activity. Al Husaini et al. (2024) demonstrated the application of deep learning models (InceptionV3, InceptionV4) to real-time thermography video for breast cancer classification, achieving promising results. Future research should

explore spatiotemporal deep learning architectures such as 3D CNNs, ConvLSTMs, and Video Transformers for exploiting the full temporal dimension of thermographic video data.

8. Comparative Summary of Thermographic Techniques

The following table summarizes the key thermographic technologies discussed in this review, their working principles, primary cancer applications, and current limitations:

Passive IR Thermography: Detects natural body heat emission; used for breast, skin, thyroid cancer; limited depth sensitivity and high environmental sensitivity.

Dynamic/Active Thermography: Applies thermal stimulus and measures tissue response; used for breast and skin cancer; requires controlled stimulation equipment and longer acquisition time.

Lock-in Thermography: Uses periodic modulated heat with lock-in detection; used for skin cancer depth assessment; complex instrumentation.

Rotational Breast Thermography: Multi-angle thermal acquisition around the breast; breast cancer; high data volume and complex reconstruction.

Infrared Thermal Modulation Endoscopy (ITME): Endoscopic thermal modulation for internal tumors; rectal/GI tumors; currently preclinical stage.

Induced Current Electro-Thermal Imaging (ICET): Magnetically induced currents to enhance thermal contrast; breast cancer; safety and depth limitations.

Smartphone-Based Thermography: Portable sensor on mobile device with AI; breast cancer pre-screening; lower sensitivity and resolution than clinical cameras.

9. Conclusion

Medical thermal imaging represents a scientifically grounded, clinically promising, and rapidly evolving approach to non-invasive cancer detection. Grounded in the thermobiological principle that malignant tumors generate excess heat through elevated

metabolic activity and angiogenesis, thermography provides a radiation-free, painless, and potentially cost-effective alternative or complement to conventional screening modalities.

This review has documented significant technological progress across multiple dimensions: from improved infrared sensor sensitivity and novel dynamic thermographic techniques to the application of sophisticated AI models including CNNs, transfer learning, and physics-informed inverse modeling. The application of thermography has expanded beyond breast cancer to include thyroid, skin, rectal, and brain tumors, broadening the potential clinical impact of the technology.

Despite these advances, major challenges persist. The field is hampered by small and non-diverse datasets, lack of standardized protocols, limited large-scale clinical validation, FDA non-approval as a primary screening tool, physiological confounders, and depth limitations for deep-seated tumors. The interpretability of AI models remains an important unsolved problem for clinical adoption.

Future research must prioritize multicenter clinical trials, standardized imaging protocols, large and diverse public datasets, federated learning for privacy-preserving AI development, multimodal fusion systems, and explainable AI frameworks. In resource-limited settings such as sub-Saharan Africa, thermography — particularly in its portable, smartphone-based form — holds particular promise for democratizing cancer screening access. As the field continues to mature, thermal imaging is poised to occupy an increasingly important role in the armamentarium of early cancer detection, complementing existing modalities and potentially enabling the detection of cancers at earlier, more treatable stages than is currently possible.

References

- Ahmed, F. (2024). Breast cancer detection with VGG16: A deep learning approach with thermographic imaging. *International Journal of Intelligent Systems and Applications in Engineering*, 12(4), 3439.
- Al Husaini, M.A.S., Habaebi, M.H., & Islam, M.R. (2024). Real-time thermography for breast cancer detection with deep learning. *Discover Artificial Intelligence*, 4, 57.
- Al Husaini, M.A.S. et al. (2025). Early breast cancer detection via infrared thermography using a CNN enhanced with particle swarm optimization. *Scientific Reports*.
- Arora, N., Martins, D., Ruggerio, D., et al. (2020). Thermal imaging: An emerging modality for breast cancer detection. *Journal of Medical Systems*, 44, Article 136.

- Bray, F., Laversanne, M., Sung, H., Ferlay, J., Siegel, R.L., Soerjomataram, I., & Jemal, A. (2024). Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, 74, 229–263.
- Chantasartassamee, P., Ongphiphadhanakul, B., Suvikapakornkul, R., Binsirawanich, P., & Sriphrapadang, C. (2024). Artificial intelligence-enhanced infrared thermography as a diagnostic tool for thyroid malignancy detection. *Annals of Medicine*, 56(1), 2425826.
- Ebsco Research. (2024). Thermal imaging. *Health and Medicine Research Starters*. Retrieved from <https://www.ebsco.com/research-starters/health-and-medicine/thermal-imaging>
- Gonzalez-Hernandez, J.-L., Recinella, A.N., Kandlikar, S.G., Dabydeen, D., Medeiros, L., & Phatak, P. (2020). An inverse heat transfer approach for patient-specific breast cancer detection and tumor localization using surface thermal images in the prone position. *Infrared Physics & Technology*, 105, 103202.
- Gutierrez, C., Owens, A., Medeiros, L., Dabydeen, D., Sritharan, N., Phatak, P., & Kandlikar, S.G. (2024). Breast cancer screening using inverse modeling of surface temperatures and steady-state thermal imaging. *Cancers*, 16(12), 2264.
- Kandlikar, S.G., Perez-Raya, I., Raghupathi, P.A., Gonzalez-Hernandez, J.-L., Dabydeen, D., Medeiros, L., & Phatak, P. (2017). Infrared imaging technology for breast cancer detection – Current status, protocols and new directions. *International Journal of Heat and Mass Transfer*, 108, 2303–2320.
- Kesztys, D., Brucher, S., Wilson, C., & Kesztys, T. (2023). Use of infrared thermography in medical diagnosis, screening, and disease monitoring: A scoping review. *Medicina*, 59, 2139.
- Kim, S. & Oh, G. et al. (2024). Infrared thermal modulation endoscopy for label-free tumor detection. *Scientific Reports*. <https://doi.org/10.1038/s41598-024-76173-8>
- Levy, A., Dayan, A., Ben-David, M., & Gannot, I. (2010). A new thermography-based approach to early detection of cancer utilizing magnetic nanoparticles: Theory simulation and in vitro validation. *Nanomedicine: Nanotechnology, Biology and Medicine*, 6(6), 786–796.
- Mamani, J. et al. (2025). Advancements in breast cancer detection: A review of global trends, risk factors, imaging modalities, machine learning, and deep learning approaches. *Journal of Imaging Informatics in Medicine*, 5(3), 46.
- Mangiacotti, M. et al. (2024). A review of techniques and bio-heat transfer models supporting infrared thermal imaging for diagnosis of malignancy. *Applied Sciences*, 14(4), 1603.
- Piotrowski, M. et al. (2025). Artificial intelligence in IR thermal imaging and sensing for medical applications. *Sensors*, 25(3), 891.
- Tanriverdi, V., & Gencer, N.G. (2024). Induced current electro-thermal imaging for breast tumor detection: A numerical and experimental study. *Annals of Biomedical Engineering*, 52, 1078–1090.
- Tewelde, F. et al. (2025). Harnessing infrared thermography and multi-convolutional neural networks for early breast cancer detection. *Scientific Reports*.
- U.S. Food and Drug Administration (FDA). (2023). Breast cancer screening: Thermogram no substitute for mammogram. *Consumer Updates*.
- Vijay, R., et al. (2022). Nodule detection in infrared thermography using deep learning. In *Proceedings of the IEEE International Conference*. IEEE Xplore. <https://doi.org/10.1109/XXXXX.2022.9824313>
- Wei, J., Tada, M., So, J., & Torres, A. (2024). Artificial intelligence and skin cancer. *Frontiers in Medicine*, 11, 1331895.
- Wilson, A.N., Gupta, K.A., Koduru, B.H., Kumar, A., Jha, A., & Cenkeramaddi, L.R. (2023). Recent advances in thermal imaging and its applications using machine learning: A review. *IEEE Sensors Journal*, 23(4), 3395–3407.
- Zhang, N., Liu, J., Jin, Y., et al. (2023). An adaptive multi-modal hybrid model for classifying thyroid nodules by combining ultrasound and infrared thermal images. *BMC Bioinformatics*, 24(1), 315.